

ENV S 193DS Final

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[Github Repo] (https://github.com/nvanderpoel-cmd/ENVS-193DS_spring-2026_final)

Set up

```
# reading in packages
library(here)
library(tidyverse)
library(janitor)
library(dplyr)

# reading in data
nests <- read.csv(here("data/nest_data_final.csv"))
```

Problem 1. Research Writing

- a. Transparent statistical methods
- b. Suggestions for rewriting
- c. Further analysis

Problem 2. Data analysis

- a. Clean and wrangle

```
# new object nests_clean
nests_clean <- nests |>
  select(height_cm, case_control, ant_species) |>
  filter(ant_species %in% c(
    "AB",
    "RRB",
    "BBR")) |>
  mutate(ant_species = as_factor(ant_species),
         ant_species = fct_relevel(ant_species,
                                   "AB",
                                   "RRB",
                                   "BBR")) |>
  mutate(species_name = case_match(
    ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
    "BBR" ~ "Crematogaster nigriceps"
  ))
```

Warning: There was 1 warning in `mutate()`.
i In argument: `species\_name = case\_match(...)`.
Caused by warning:
! `case\_match()` was deprecated in dplyr 1.2.0.
i Please use `recode\_values()` instead.

```
slice_sample(nests_clean, n = 10)
```

	height_cm	case_control	ant_species	species_name
1	175.0	0	RRB	Crematogaster mimosae
2	65.0	0	BBR	Crematogaster nigriceps
3	118.0	0	RRB	Crematogaster mimosae
4	237.5	0	RRB	Crematogaster mimosae
5	208.0	0	RRB	Crematogaster mimosae
6	391.5	1	RRB	Crematogaster mimosae
7	236.0	0	RRB	Crematogaster mimosae
8	164.5	0	BBR	Crematogaster nigriceps
9	513.0	0	RRB	Crematogaster mimosae
10	123.0	0	RRB	Crematogaster mimosae

b. Response variable

For the response variable, case_control, the 1's signify there is a nest, of any bird species, present, while a 0 signifies there is no nest present.

c. Purpose of study

Tree height: Tree height is a continuous variable. As tree height increases, the probability of a nest occurrence would increase because of increased distance between nests and browsing mammals.

Acacia ant species: Acacia ant species is a categorical variable.

d. Table of models

Model number	Nest?	Tree Height	Ant Species	Interaction
1	X	X	X	X
2	X	X	X	

- e. Run the models
- f. Select the best model
- g. Check the diagnostics
- h. Visualize the model predictions
- i. Write a caption for your figure
- j. Calculate model predictions
- k. Interpret your results

Problem 3. Affective and exploratory visualizations

- a. Comparing visualizations
- b. Designing an analysis
- c. Check any assumptions and run your analysis
- d. Create a visualization
- e. Write a caption
- f. Write about your results
- g. Sharing your affective visualization